

An automatic tracking method for fruit abscission of litchi using convolutional networks

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ABSTRACT

Automatic tracking of litchi fruit abscission provides important information for the orchard management. Currently, the prevalent method for tracking litchi fruit abscission involves manual selection of branches followed by daily fruit counting, a process deemed both inefficient and subjective. Therefore, this paper proposed a novel approach that integrated an improved image matching and object detection algorithms to intelligently analyze litchi fruit abscission of branches in the orchard. The method consisted of three steps. Firstly, an automatic fixed-point focus system was developed to capture long-term image sequences of selected branch areas. From April 27th to June 21st, 2022, the system autonomously navigated to specified locations and automatically adjusted the focal length to capture daily images of selected branches, thereby forming image time series is crucial for analysis. Secondly, to reduce fruit counting errors due to interferences from other branches, an improved SuperGlue algorithm named SE2-SuperGlue was developed to match branches and extract the region of interest of the target branch (ROI-B) from the image time series. The SE2-SuperGlue algorithm introduced an equivariant steerable CNN that is equivariant to translations and image rotations, rendering it robust to rotations and ensuring accurate matching amid changes in branch angles. Thirdly, the ROI-B sequences were input into the YOLOv8n to detect litchi fruits, yielding tracking results of litchi fruit abscission on the image time series. Experimental results showed the good performance of the proposed method. The SE2-SuperGlue network achieved higher matching score (0.53) and a speed of 177 ms/frame compared to conventional matching algorithms, successfully extracting the ROI-B sequence. Additionally, YOLOv8n achieved an mAP@0.5 of 98.7 %, and the correlation coefficient R^2 between prediction and ground truth for automatic tracking of daily fruit count over a 56-day image time series of selected branches was 0.98. Furthermore, the study analyzed a continuous 56-day image sequence depicting relative fruit abscission for 80 branches of three varieties (Guiwei, Nuomici, and Huaizhi), along with the peaks of fruit abscission.

1. Introduction

China boasts the largest cultivation area and highest litchi output globally, making it a pivotal player in the litchi industry. Litchi serves as a significant economic crop in southern China, where it enjoys widespread cultivation (Qi et al., 2019). Despite its prominence, litchi yields remain suboptimal, largely due to the prevalent occurrence of litchi fruit abscission. It has been widely documented that litchi fruits undergo three distinct peak periods of abscission prior to maturation, each attributed to varying factors, which contributes to the inconsistent and

often low litchi yield, as well as the occurrence of alternating “on” and “off” years (Li et al., 2019; Zhao et al., 2020). The study of litchi fruit abscission is crucial for industry management and market forecasting, providing essential information for farmers, businesses, and the government. This, in turn, promotes the stable development of the litchi market. Moreover, it holds significant implications for regulating seed and fruit production, as well as enhancing breeding and harvesting practices (Estornell et al., 2013; Yi et al., 2023). The fast and accurate monitoring of litchi fruit abscission dynamics facilitates the timely implementation of fruit preservation strategies, thereby contributing to

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(a) The image of 1st day's branch(b) The image of 22nd day's branch

Fig. 1. Illustration for shooting branch. The yellow circle indicates adjacent branches are captured. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the improvement of yields. To obtain precise data on fruit abscission, researchers typically select and tag representative inflorescence branches, which are then tagged. Subsequently, from the third week after the flowering season until harvest, the number of fruits on the tagged branches is manually counted every seven days to analyze the relative fruit abscission rate (Fan et al., 2015; Yan et al., 2019). This is undoubtedly labor-consuming, and the efficiency of statistical analysis is constrained by factors such as orchard environment, canopy structure, and weather conditions.

Currently, deep learning has been widely applied in the field of fruit detection and counting (Gao et al., 2020; Wu et al., 2021; Wang et al., 2022). These detection algorithms can be broadly categorized into two types: region proposal-based target detection methods, such as Mask R-CNN (He et al., 2017) and Faster R-CNN (Ren et al., 2017), and regression-based target detection, such as SDD (Liu et al., 2016) and the YOLO series (Redmon et al., 2018; Jocher et al., 2021; Wang et al., 2023). In contrast, regression-based target detection algorithms have faster detection speeds while ensuring detection accuracy, and are more suitable for agricultural applications. Among them, single-stage networks, represented by YOLO (You Only Look Once), hold a significant

advantage in agricultural detection scenarios due to their rapid detection speed and superior accuracy. Liu et al. (2020) replaced the traditional rectangular bounding box of the YOLOv3 model with a circular one for tomato detection, achieving a 96 % mean average precision (mAP) and a detection speed of 0.054 s on a 3648×2056 pixel image. Lyu et al. (2022) proposed a lightweight citrus detection network, YOLOv5-CS (Citrus Sort), to achieve accurate detection and counting of green citrus in natural environments, with a mAP of 98.23 %. Wang et al. (2022) detected ripe litchi in the orchard based on an improved YOLOv5 model, and the improved model's mAP presented a 3.5 % improvement. In summary, existing studies on fruit detection and counting have mainly focused on targets with colors that are highly distinguishable from the background such as tomatoes (Das et al., 2020), oil palms (Ali et al., 2020) and citrus fruits (Chen et al., 2022).

However, the precise counting and tracking of litchi fruit abscission of branches still encounter two major challenges. Firstly, while capturing images on a large scale and counting can be used for statistical analysis of litchi fruit abscission, the relatively small size of litchi targets in the large-scale images, coupled with the similarity in color between the green litchi fruit and the surrounding area, often lead to lower

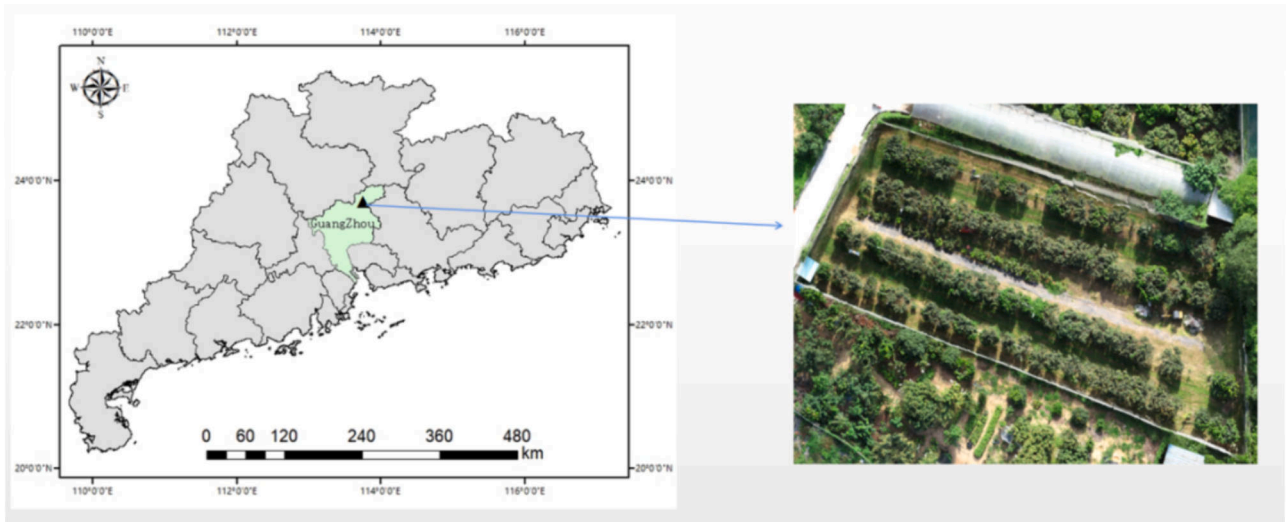


Fig. 2. The location of litchi orchard.

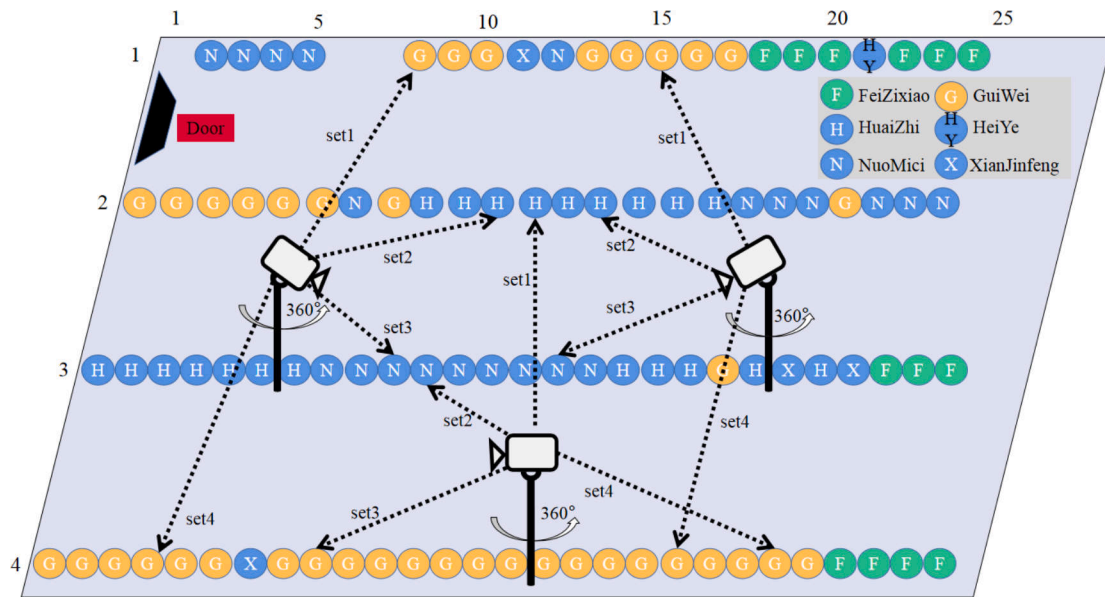


Fig. 3. The distribution of litchi trees in the orchard. The angles of cameras can rotate to a fixed position in both horizontal and vertical directions, such as set1, set2, set3 and set4, to capture images.

counting accuracy of litchi fruit. Therefore, long-term tracking and imaging of target branches are more conducive to physiological analysis of litchi fruit abscission. Furthermore, achieving such tracking and imaging in complex orchard environments poses challenges. Previous fruit studies have primarily relied on smartphones or cameras for data acquisition (Afonso et al., 2020; Vasconez et al., 2020; Liang et al., 2020). It is difficult to control the angles and distances of these devices, which makes it challenging to automatically capture high-quality images of target branches over extended periods. Secondly, precisely localizing the target branch region within a dynamically changing branch to minimize potential interference from the fruits of neighboring branches remains a substantial challenge. In natural conditions, the growth environment of litchi tree is complex, characterized by closely connected branches that undergo significant morphological changes over certain growth periods. When capturing images of target branch, in order to capture the area of the entire branch for a long time, the camera's field of view must exceed the branch area to prevent the branch from moving out of frame due to factors such as wind, rain, and

growth processes causing bending. But, with a larger shooting area, other branches may also be captured simultaneously, leading to interference, as indicated by the yellow circle in Fig. 1. Interference from neighboring branches was present when the initial image area was established, but not severe (See Fig. 1a). Due to the natural environment and growth and other factors, the interference from other branches in the image taken at the same angle was more severe (See Fig. 1b). Therefore, it is necessary to extract the region of interest of the branch (Hereinafter referred to as ROI-B) from long-term acquisition of time-series images.

To extract ROI-B (See Fig. 15), image matching (Schönberger et al., 2017; Luo et al., 2018; Balakrishnan et al., 2018;) can be used to identify and align the contents or structures with identical or similar attributes between two images at the pixel level (Ma et al., 2021). Traditional algorithms are mainly based on SIFT features (Sun et al., 2023) and SURF features (Cui et al., 2023). However, these methods often rely on manual prior knowledge to extract feature keypoints and design complex handcrafted descriptors, leading to high computational requirements



Fig. 4. The real camera system in the orchard. The yellow circle indicates the cameras used in this research, the other cameras were installed after this research and are not related to this research. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and low matching accuracy and limited real-time performance. In recent years, deep learning has become the dominant approach to improve image matching performance. Savinov et al. (2017) proposed Quad-Networks for keypoint detection, which employed unsupervised learning to train a neural network and achieved robust matching for remote sensing images. Hoffmann et al. (2019) utilized a fully convolutional network to calculate the degree of similarity or correlation between features. DeTone et al. (2018) utilized the Superpoint network to extract keypoints and employed the (Sarlin et al., 2020) network for feature point matching. By leveraging end-to-end feature point matching, the SuperGlue algorithm overcomes poor performance due to significant disparities and distorted transformations. Deep learning-based image matching algorithms have been widely applied in various fields such as medical imaging and heterogeneous image analysis. Nonetheless, there has been limited research on deep learning-based image matching algorithms for orchard research. Changes in branch angles in orchard scenes due to factors such as wind, rain, and growth, the SuperGlue algorithm is sensitive to changes in angles, affecting matching accuracy.

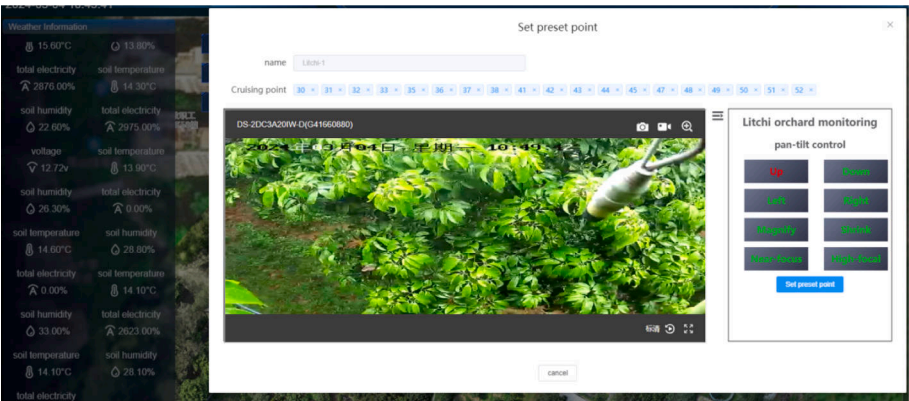
Overall, this study introduces an innovative method for the automated tracking of litchi fruit abscission rates on target branches in orchards, through the integration of object detection and image matching techniques. Firstly, a fixed-point and fixed-focus capture system was developed for autonomous long-term (56 days) tracking and imaging of target branches. Secondly, the improved SuperGlue algorithm was employed for image matching of branches to obtain the sequences of ROI-B from the image time series of target branches. To address the matching algorithm's sensitivity to angle changes, an equivariant steerable CNN, invariant to translations and rotations, was introduced to enhance the SuperGlue algorithm's robustness to rotations. Finally, the sequence of ROI-B were input into the YOLOv8 object detection network

to generate long-term sequential statistics on litchi fruit abscission. This research can provide reliable foundational data for fruit preservation, fertilization, and other aspects of orchard management.

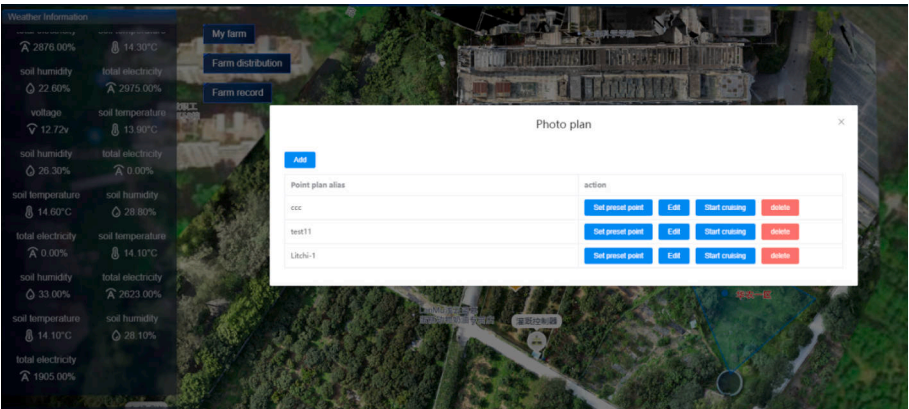
2. Materials and methods

2.1. Image acquisition using fixed-point focus system

The RGB images used in this study were captured at the SCAU Main Campus Teaching & Research Base located in Guangzhou, Guangdong Province, China ($23^{\circ}1'58.29''N$, $113^{\circ}3'59.93''E$) (See Fig. 2). The panoramic layout of the orchard and fruit trees is depicted in Fig. 3. Within the experimental orchard, the spacing between rows of fruit trees measures 6 m, while the gap between individual trees is 2.5 m. The predominant varieties of fruit trees in the orchard include Guiwei, Nuomici, and Huaizhi, with Guiwei trees approximately double in number compared to Nuomici and Huaizhi varieties, respectively. The height and canopy size of each fruit tree within the orchard exhibit variation. To monitor the growth of litchi and obtain experimental images timely, three Hikvision DS2DC3A20IW-D (4-megapixel, image size of 2560×1440 pixels) zoom cameras were installed in the orchard and are capable of automatic rotation (See Fig. 4). Positioned at a height of 6 m above the ground, these cameras feature an IP66 level of protection, offering waterproof, dustproof, and corrosion-resistant properties suitable for diverse outdoor environments. Additionally, the cameras are equipped with a $23 \times$ zoom function, enabling manual adjustment of focus and angle for long-distance or close-up photography and monitoring from various distances and perspectives. The distance between the cameras and the branches was approximately 1 m at the closest and 30 m at the farthest. For automated fixed-point and fixed-focus image capture, this research has developed a data acquisition system based on



(a) The setting interface of preset points



(b) The execution interface of shooting task

Fig. 5. The platform of image acquisition.

Table 1
The experimental dataset.

	Train	Test	Total
Image matching	2704	676	3380
Object detection	2400 (119,452 labels)	150 (7635 labels)	2550 (127,087 labels)

the Windows 10 operating system, Python 3.7 compilation environment, OpenCV image processing library, and PyQt5 toolkit. The platform interfaces with Hikvision's Ezviz cloud to control camera operations. The system records angle and focal length information for multiple manually preset sampling points, subsequently scheduling patrols of these preset points at designated times to execute corresponding shooting tasks automatically (See Fig. 5). Each camera can preset up to 300 shooting points. Using fewer cameras to capture images from multiple acquisition points. Thus, automatic long-term tracking and imaging of target branches are achieved.

In this study, the horizontal 360° and vertical 120° rotation of the cameras were manually controlled through a data acquisition system (See Fig. 5a), and randomly selected 10 Guiwei trees to set 40 points, 5 Nuomici trees with 20 points, and 5 Huaizhi trees with 20 points, respectively, similar to the methods used in Hu et al. (2020) and Ma et al. (2021). For each preset point, the camera focus was manually adjusted to ensure clear images capture. Leveraging the recorded angle and focal length of the preset points, our acquisition platform facilitated daily camera navigation and image capture of the preseted branches

(See Fig. 5b). The camera was controlled to cruise once daily, resulting in continuous image sequences of 80 branches over a span of 56 days. All images were captured between 8 am and 6 pm from April 27th to June 21st, 2022, encompassing diverse lighting conditions (strong, moderate and weak light), scales (zooming in or out of the shooting range), and weather conditions (rainy, cloudy, sunny, etc.) These images would be used for tracking litchi fruit abscission. In addition, we randomly captured images of other branches using the system as training and testing sets for matching and detection algorithms.

2.2. Building the dataset

The dataset used in this study consisted of two parts: image matching dataset and object detection dataset. Table 1 provides detailed quantities for each dataset. The image matching dataset in this study comprised 3755 original images of litchi fruits. The 3755 images were independent of the images of the selected 80 branches. These images (3755) were randomly selected and captured manually through the data acquisition system during the growth process of branches. Due to factors such as shooting angle and lighting conditions, some of the collected images have underexposure and poor clarity. Therefore, it was necessary to filter out the images that did not meet the training requirements. After filtering, 3380 appropriate images were selected from the original dataset. The experimental dataset was then randomly split into training and testing sets in an 8:2 ratio, with 2704 images used for training the image matching network and the remaining 676 images used for testing.

From the previous pool of 3755 images, a subset of 750 green litchi images was selected as the object detection dataset, comprising samples

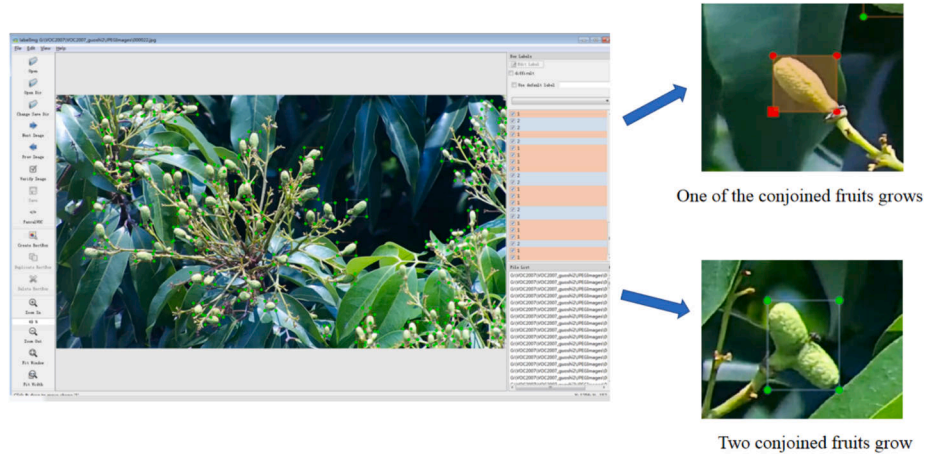


Fig. 6. Example of data annotation.

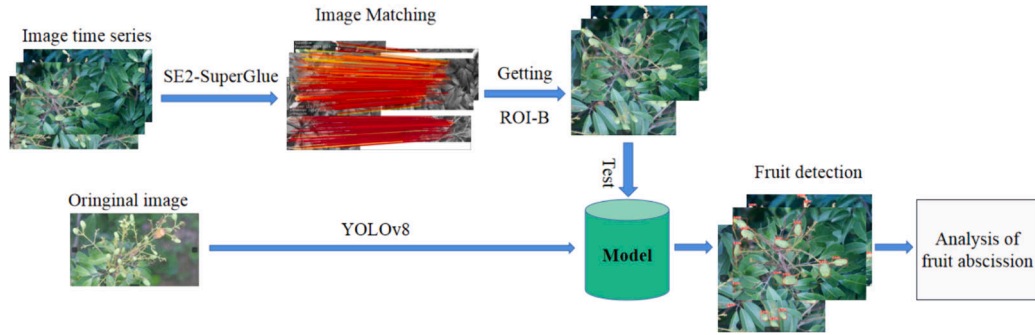


Fig. 7. Flow chart of the proposed method.

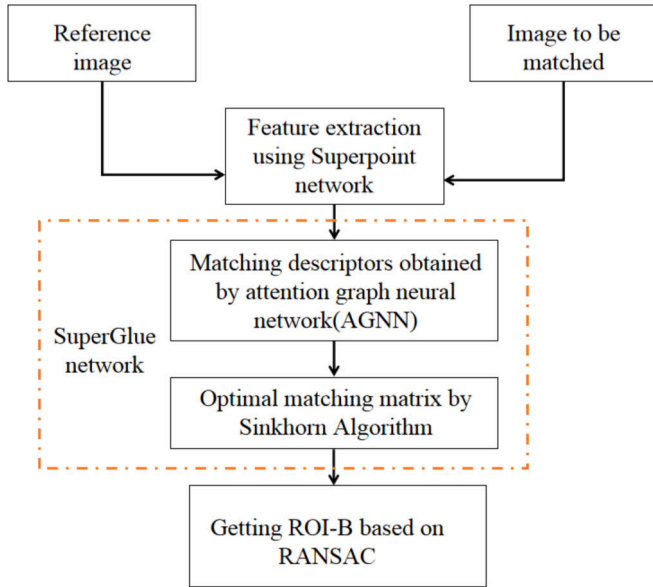


Fig. 8. Flow chart of litchi branch image matching algorithm.

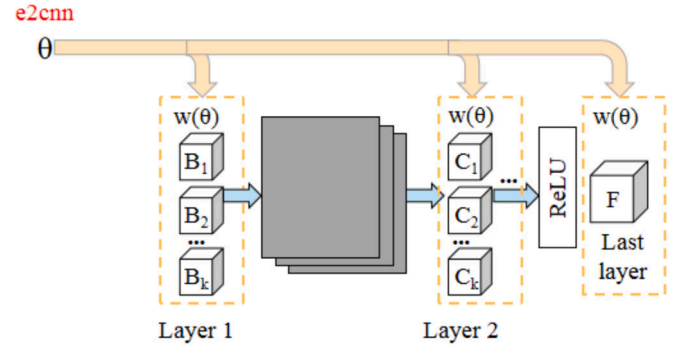


Fig. 9. The e2cnn architecture.

with diverse poses, sizes, backgrounds, densities, and occlusions. According to the biological characteristics, litchi fruits can be classified into two categories: class 1 (One of the conjoined fruits grows) and class 2 (Two conjoined fruits grow) (Wei et al., 2013). These were labeled using the Labeling script, as illustrated in Fig. 6. On average, each image contained 5–91 fruits, resulting in a dataset with 37,516 labels. The

proportion of the two categories was approximately 1:1.40, indicating no significant data imbalance. Subsequently, the annotated dataset was randomly divided into training and testing sets with an 8:2 ratio. The training set image was augmented using image enhancement methods such as vertical and horizontal mirroring, displacement, blur, and rotation to enrich feature learning. A total of 2400 images contained 119,452 labels were used for network training. The test set consisted of 150 images and 7635 labels. Ultimately, the object detection dataset contained 2550 images data and 127,087 labels.

2.3. Counting litchi fruit on branches in the image sequence

The proposed method is illustrated in Fig. 7. Initially, to reduce the impact of neighboring branches on fruit counting for a selected branch, an improved image matching network was proposed to acquire the ROI-

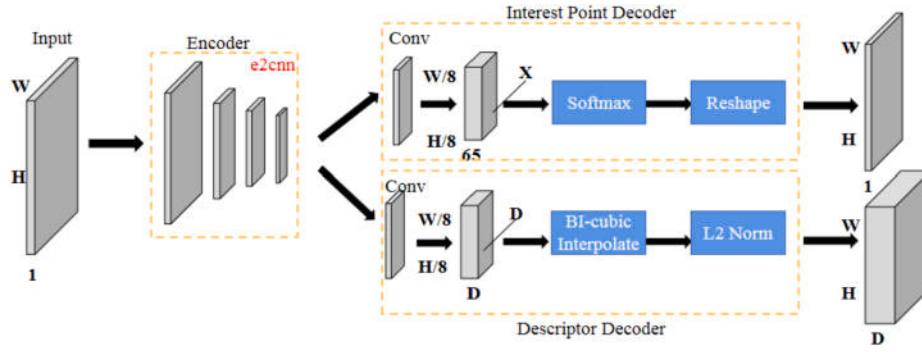


Fig. 10. The improved SuperPoint architecture.

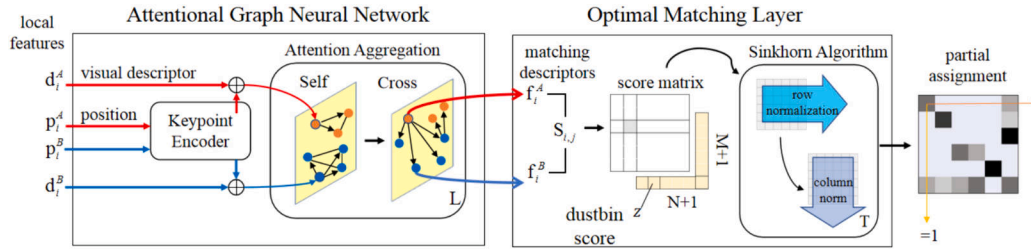


Fig. 11. The SuperGlue architecture.

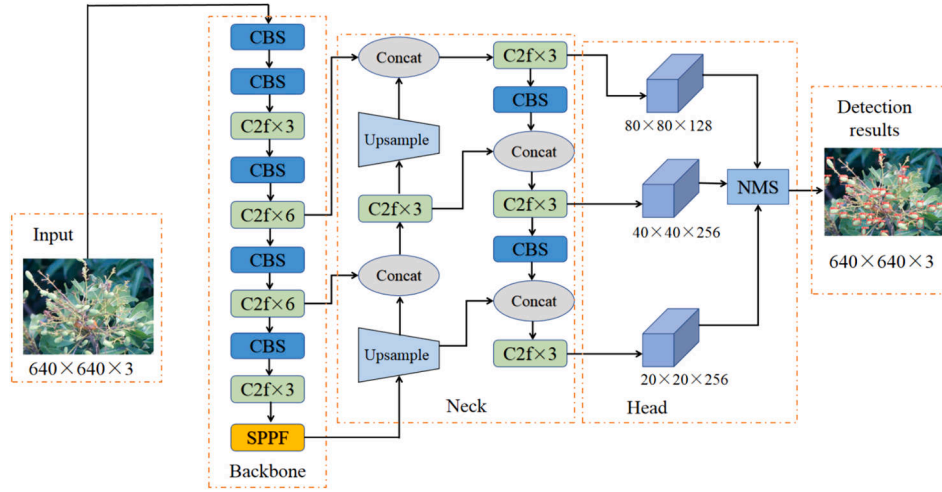


Fig. 12. The structure of YOLOv8 model.

Table 2

Comparison between SE2-SuperGlue and other algorithms.

Local features	Matcher	Features num	Score (↑)	Speed (ms/f)
ORB	KNN	1024	0.14	275
SURF	KNN	1024	0.20	393
SIFT	KNN	1024	0.28	480
VGG + SuperPoint	SuperGlue	1024	0.46	103
e2cnn + SuperPoint (ours)	SuperGlue	1024	0.53	177

B sequences. Subsequently, the obtained ROI-B sequences were fed into a trained YOLOv8 network for fruit detection. Finally, the fruit abscission of the 80 branches was counted and analyzed.

2.4. Improved image matching network for ROI-B sequence extraction

The flowchart of the SuperGlue algorithm for litchi branch image matching is shown in Fig. 8. The algorithm primarily comprises two components: Inputting a pair of images with certain similarity, utilizing an improved SuperPoint network for feature extraction (detection and description) and the SuperGlue network for feature matching and filtering. Using the filtering results derived from the SuperGlue network, the homography matrix is calculated, and finally, the ROI-B is obtained.

2.4.1. E(2) equivariant steerable CNNs

This research employed the framework of E(2) equivariant steerable CNNs (Weiler et al., 2019) to construct dense descriptors. Steerable CNNs represent examples of group equivariant neural networks. A fundamental property of CNN layers is their translation equivariance, where if T represents a translation, L a CNN layer and I an image, then $L(T(I)) = T(L(I))$. This property enables CNNs to process image features

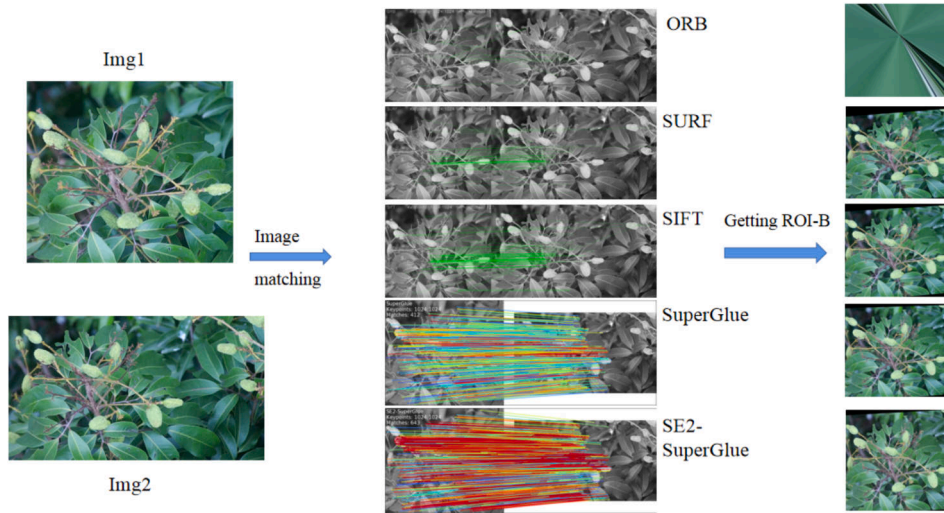


Fig. 13. Visualization for comparison between SE2-SuperGlue and traditional feature matching algorithms.

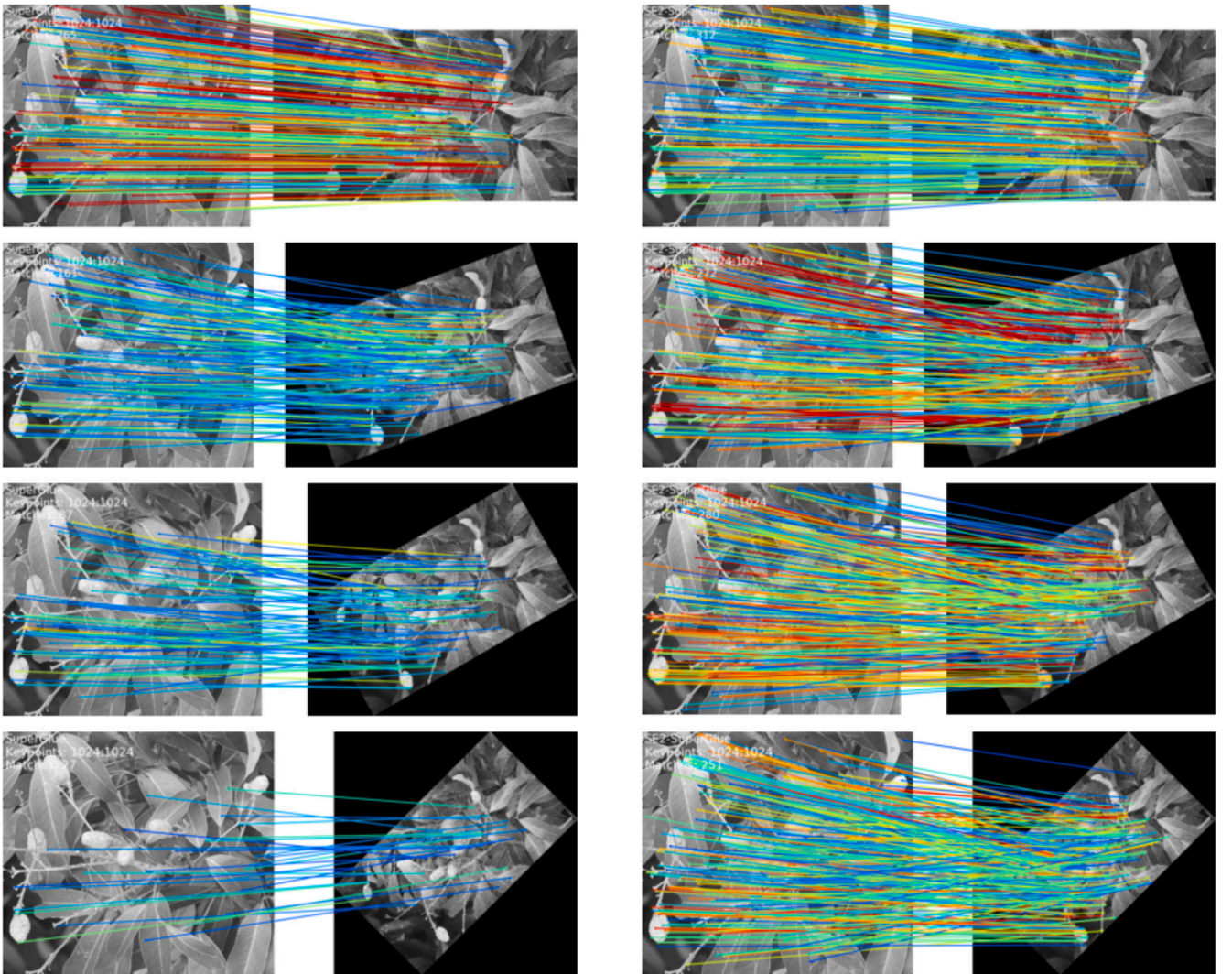
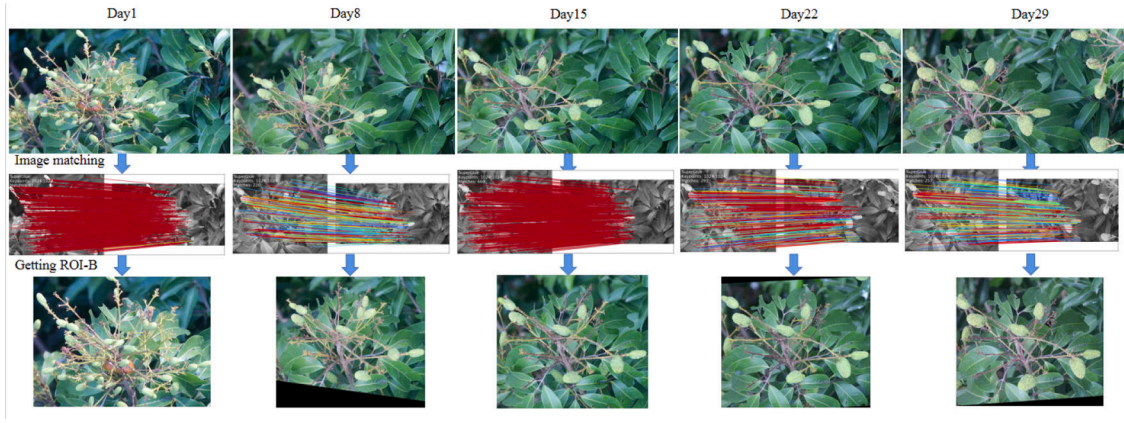
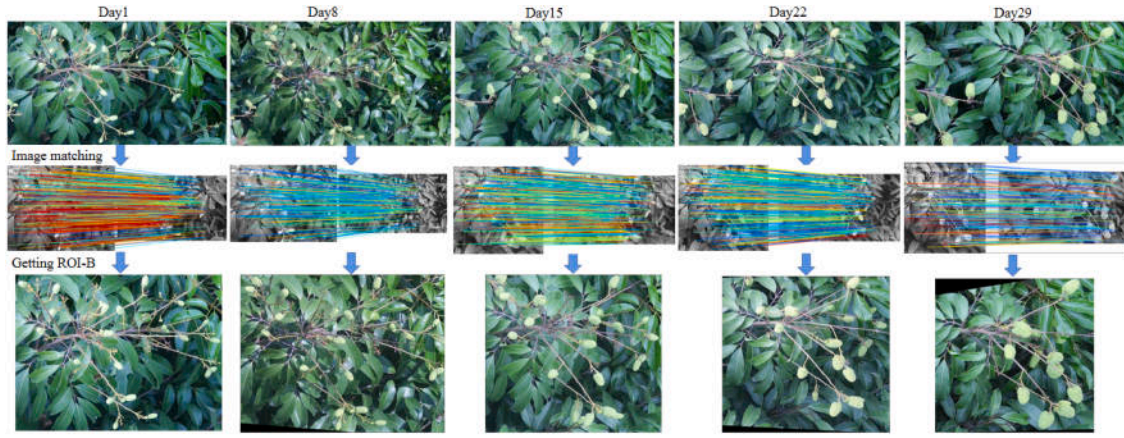


Fig. 14. Left: SuperGlue. Right: Our SE2-SuperGlue. The bottom three pairs are augmented by inplane rotations of 20°, 30° and 45°, respectively.



(a) Visualization of image matching for branch a



(b) Visualization of image matching for branch b

Fig. 15. The illustration of image matching for two branches (a and b).**Table 3**

Performance on comparison of different algorithms.

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Model Size (Mb)	Speed (ms/f)
YOLOv3-tiny	86.7	54.5	17.42	100.64
YOLOXs	89.7	68.4	71.85	72.78
Faster R-CNN	91.0	71.8	165.76	21.58
YOLOv5s	98.3	80.5	16.81	114.37
YOLOv8n	98.7	84.0	6.20	176.25

uniformly, irrespective of their position within an image. Extensive research has focused on describing neural networks that exhibit equivariance under more general groups beyond translations (Cohen et al., 2019; Lang et al., 2020; Weiler et al., 2021; Aronsson, 2022). Under mild assumptions, linear layers demonstrating group equivariance are necessitated to adopt a convolutional form. The theory of linear equivariant layers is formulated in terms of *representation theory* (Zimmermann, 2014).

The equivariant steerable CNN composes of a number of steerable convolutional layers consisting of multiple steerable filters, as illustrated in Fig. 9 (Li et al., 2020). This steerable CNN has a steering parameter θ that can be used to make it sensitive to the parts that have orientation θ . The steerable filters are obtained as a linear combination from a basis B_1 ,

$B_2, \dots, B_k, C_1, C_2, \dots, C_k$ and F , and can be rotated to any angle θ by a simple re-weighting of the basis. The response maps of each layer for any angle θ can be obtained by convolution with linear combination of all filters and weight vector $w(\theta)$. The Steerable CNN for this task consists of 4 steerable convolutional layers. The third steerable convolutional layer is followed by ReLU activation, and the last one returns the response. We used the feature extraction method of *e2cnn* in backbone to solve the problem that SuperGlue is sensitive to angle changes.

2.4.2. SuperPoint features extraction

The growth of litchi branches, along with factors like rain and wind, causes changes in branch morphology and angle. SuperPoint's feature extraction relies on convolutional operations, which are highly sensitive to angle changes. This sensitivity significantly impacts the accuracy of image matching and poses challenges in accurately extracting the ROI-B from the branch image time series. To address this issue, this research proposes an improved SuperGlue algorithm using equivariant steerable CNNs to enhance its robustness to rotations.

The improved SuperPoint is a deep learning method for feature point detection and descriptor extraction. The network (See Fig. 10) consists of encoder, interest point decoder and descriptor decoder. The network firstly uses a VGG-Style (Simonyan and Zisserman, 2014) encoder downsize the image and extract features. This encoder section comprises a convolutional layer, a Max-Pooling layer and a non-linear activation layer. We substituted the CNN backbone of the encoder with equivariant steerable CNNs (Weiler et al., 2019), and called the modified models

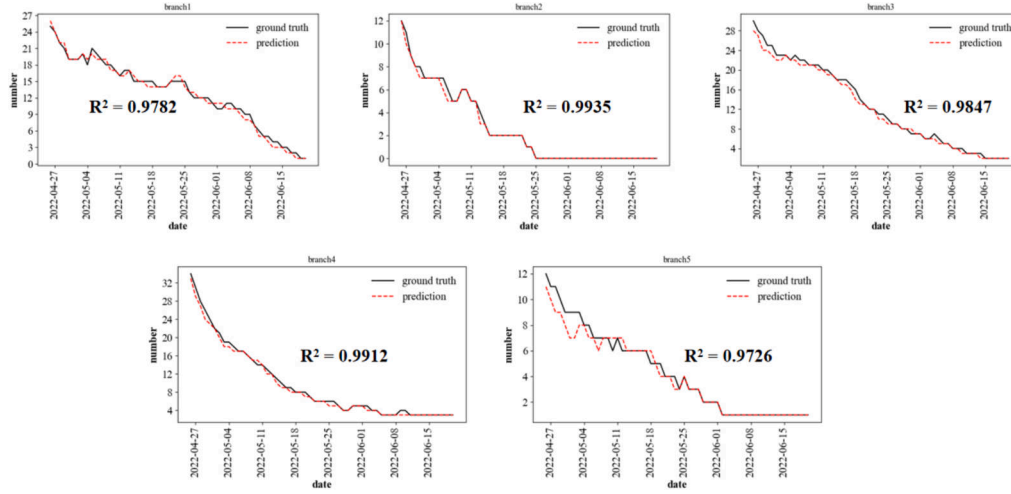


Fig. 16. The curves of 5 branches between the ground truth and the predicted values.

SE2-SuperGlue. The encoder structure of the two convolutional neural networks is same, with shared parameters. The interest point decoder convert the feature maps outputted by the encoder into a heatmap of size $H \times W \times 1$, outputting the positions and confidence of the feature points. Meanwhile, the descriptor decoder normalizes the semi-dense descriptor of $(H/8) \times (W/8) \times D$ to a dense descriptor of size $H \times W \times D$.

The improved SuperPoint operates on a self-supervised training mechanism, eliminating the need for manual annotation. The algorithm inputs an image and trains to extracts feature points and descriptors. Then, it uses trained feature point decoders and descriptor decoders to generate a new image corresponding to the feature points based on random homography transformations, forming a pair of image pair. The feature points and descriptors of this image pair will be used as input to the SuperGlue network.

2.4.3. SuperGlue features matching

SuperGlue is a network designed to execute feature matching and outlier removal simultaneously. Feature matching is achieved by solving a differentiable optimal transport problem, and the loss function is constructed using a Graph Neural Network (GNN). The SuperGlue network, depicted in Fig. 11, consists of two main modules: the attention-based GNN (AGNN) and the optimal matching layer (Sarlin et al., 2020). AGNN maps the coordinates of the keypoints p and the visual descriptors d obtained through the SuperPoint network into a vector f (feature matching vector) using a keypoint encoder, which utilizes convolution to obtain local feature points. The feature matching performance of vector f is enhanced through Self-attention and Cross-attention (repeated $L = 7$ times).

Subsequently, the feature matching descriptors f_i^A and f_i^B obtained from AGNN are further multiplied by inner product to obtain the score matrix S_{ij} , which is used to calculate the similarity between the descriptors in images A and B . The final allocation matrix P is obtained by maximizing the overall score $\sum_{i,j} S_{ij} P_{ij}$. However, due to the absence of matching points for certain feature points, a dustbin channel is added as an additional row and column on the score matrix to filter out incorrect matching points, resulting in an augmented matrix S . Then, the Sinkhorn algorithm is used to maximize $\sum_{i,j} S_{ij} P_{ij}$ and obtain the optimal matching matrix P .

As designed, both the GNN and the optimal matching layer in SuperGlue are differentiable, enabling backpropagation from matching to visual descriptors. SuperGlue is trained in a supervised manner from the ground truth matches $M = \{(i, j)\} \subset A \times B$, which are estimated using real relative transformations. If the keypoint $I \subseteq A$ and $J \subseteq B$ have no reprojections in each other's neighborhoods, they are labeled as non-matches and the negative logarithmic likelihood of the assignment

matrix P is minimized.

2.4.4. ROI-B acquisition

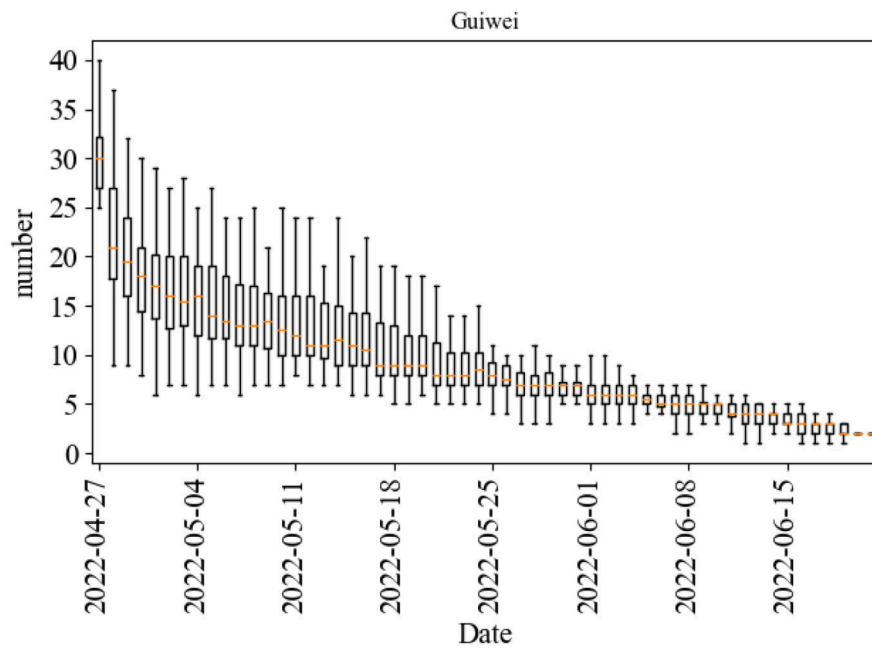
Once the position information of matched keypoints is obtained from the SuperGlue network, a robust RANSAC-based method is employed to calculate the projective transformation matrix H between imgA and imgB. The input, m_1 and m_2 for imgA and imgB, respectively, are two corresponding sequences of keypoint coordinates, where each pair of data corresponds to one another. These sequences contain both correct matches, referred to as inliers, and incorrect matches, referred to as outliers. The RANSAC method aims to separate the correct matches from the data that includes incorrect matches and estimate the homography matrix H . Subsequently, the matrix H is used to extract ROI-B based on Perspective Transformation.

2.4.5. ROI sequence of target branches

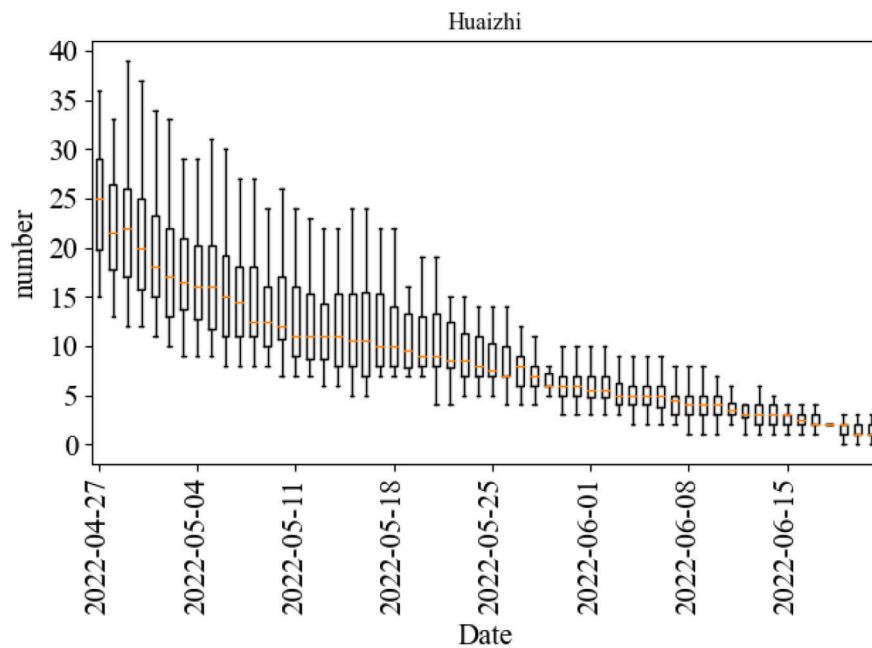
Based on the survey, on April 27th, 2022, the Guiwei, Nuomici, and Huaizhi varieties were at 17th, 17th, and 18th day after flowering, respectively. This paper selected April 27th as the first day of fruit abscission. The high variability of the litchi growth process led to more pronounced differences in branch morphology over a 56-day period. According to calculations, the ROI-B area reached around 31.6 %~70.2 % of original image area. For each branch, images of branches on the 10th, 28th, and 46th days were selected and manually cropped out of the ROI-B for image matching, respectively. Specifically, the algorithm matches the image from the 10th day with the images captured on days 1–19, the image from the 28th day with the images captured on days 20–35, and the image from the 46th day with the images captured on days 36–56.

2.5. Litchi fruit detection network

As the latest version of the YOLO series, YOLOv8 inherits the advantages of previous versions and further optimizes speed and accuracy. It performs well on various benchmark tests and datasets. YOLOv8 provides a user-friendly API and documentation, making it easier for developers to integrate and use the model. In addition, the YOLO series has always been popular as it is easy to train and tune features. The YOLOv8 model consists of four parts: Input, Backbone, Neck, and Head. The input layer adopts the mosaic data augmentation method and incorporates adaptive adjustment functions for anchors and images. In the backbone section, YOLOv8 largely mirrors YOLOv5, but replaces the C3 module with the C2f module. Drawing inspiration from the ELAN concept in YOLOv7, the C2f module combines C3 and ELAN, enhancing YOLOv8's ability to capture comprehensive gradient flow information.



(a)



(b)

Fig. 17. Variation of fruit set over 56 days. a, b, c: the distribution of fruit set of all branches per day for Guiwei, Huaizhi, and Nuomici.

The widely used SPPF module is retained at the end of the Backbone. This spatial pyramid pooling module, with 3 pooling cores of 5×5 , 9×9 , and 13×13 , extends the receptive field without compromising computation speed. The neck network continues to adopt the PAN-FPN structure, focusing on generating feature pyramids. By integrating bottom-up and top-down pathways, it facilitates information aggregation, allowing each level to capture more detailed features and improving the model's object detection capability across various scales.

Significant changes in the Head component include transitioning from the original decoupled head to the coupled head, and from Anchor Based in YOLOv5 to Anchor Free. YOLOv8 adopts CIOU and Distribution Focal Loss as the loss function for bounding box regression.

According to the complexity of the network, the YOLOv8 model is divided into versions such as x, 1, m, s, and n. To balance accuracy and speed, YOLOv8n is chosen in this study. The YOLOv8 network is shown in Fig. 12.

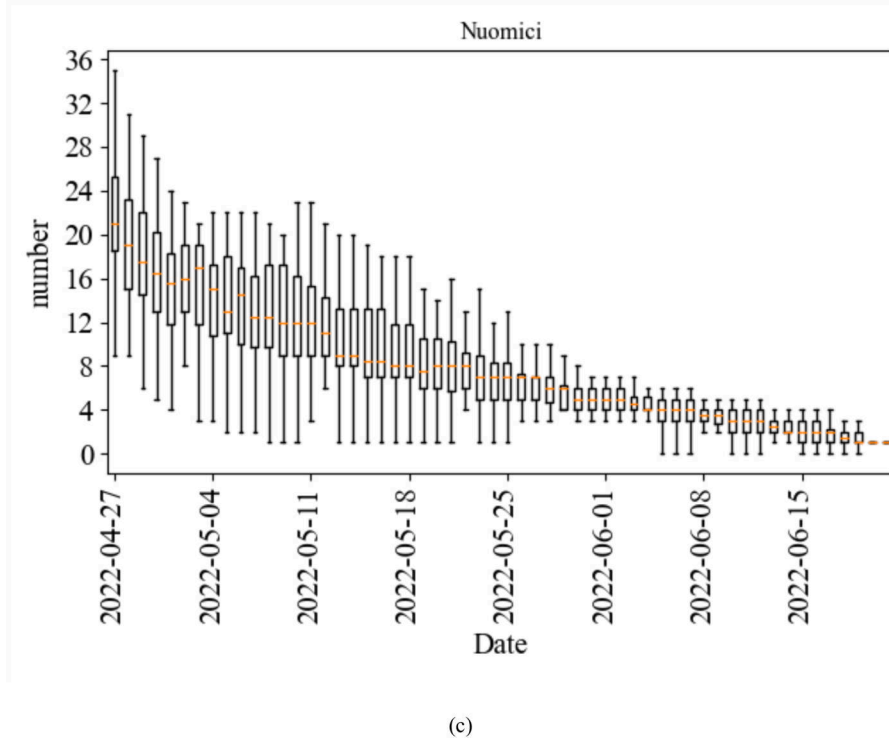


Fig. 17. (continued).

3. Experimental results and analysis

3.1. Experiment platform

The training environment used in this study includes Ubuntu 18.04, 64-bit Linux operating system, 32 GB RAM, and an NVIDIA GeForce RTX TITAN graphics card. The PyTorch-based code was employed to train the SE2-SuperGlue and YOLOv8 models from scratch using pre-trained weights provided by the SuperGlue and YOLOv8 official repositories.

The SE2-SuperGlue network was trained and its effectiveness was verified on the test set. An input image size of 640×480 was chosen for the model, with Adam serving as the optimizer and an initial learning rate set to 0.0001. The keypoint confidence threshold was set to 0.01, with an NMS radius of 8, a matching threshold of 0.3, and 50 Sinkhorn iterations. The model was trained for 10 epochs, with a default batch size of 1. The weights of convolutional layers of the equivariant Steerable CNN were initialized from the normal distribution with standard deviation 0.01. For the equivariant Steerable CNN, we started with a learning rate of $1 \cdot 10^{-6}$. After every epoch the learning rate was multiplied by 0.8. Pretrained weights were used to train the model from scratch, which took approximately 120 h. Codes are available at <https://github.com/sunset326/SE2-SuperGlue>.

This study trained the YOLOv8 network, utilizing a 640×640 input size for the model. Adam was employed as the optimizer with an initial learning rate of 0.01 and a weight decay of 0.0005. The model underwent training for 300 epochs with a batch size of 16.

3.2. Evaluation method

In this study, the matching score (Mikolajczyk et al., 2005) and the average processing time per frame (in milliseconds) on the test set were served as performance metrics for evaluating the proposed algorithm. The matching score, defined as the proportion of correctly matched feature points to the total number of matches, is given by Eq. (1).

$$\text{score} = \frac{\sum_{i=1}^Q \frac{\# \text{correct matches}}{\# \text{features num}}}{Q} \quad (1)$$

where *correct matches* represents the number of correctly matched feature points, *features num* represents the total number of extracted feature points, and *Q* represents the number of test datasets. The higher the score, the better.

This research used the most commonly applied evaluation metrics for object detection models, including mean average precision (mAP), model size (weight), and frames per second (FPS). The higher the FPS, the faster the model's detection speed, while mAP is the average of multiple AP scores and reflects the general detection performance of the algorithm for all classes, which is a more comprehensive metric to evaluate the overall performance of the network. The calculation formulas are shown in Eqs. (2)–(5).

$$P = \frac{TP}{TP + FP} \quad (2)$$

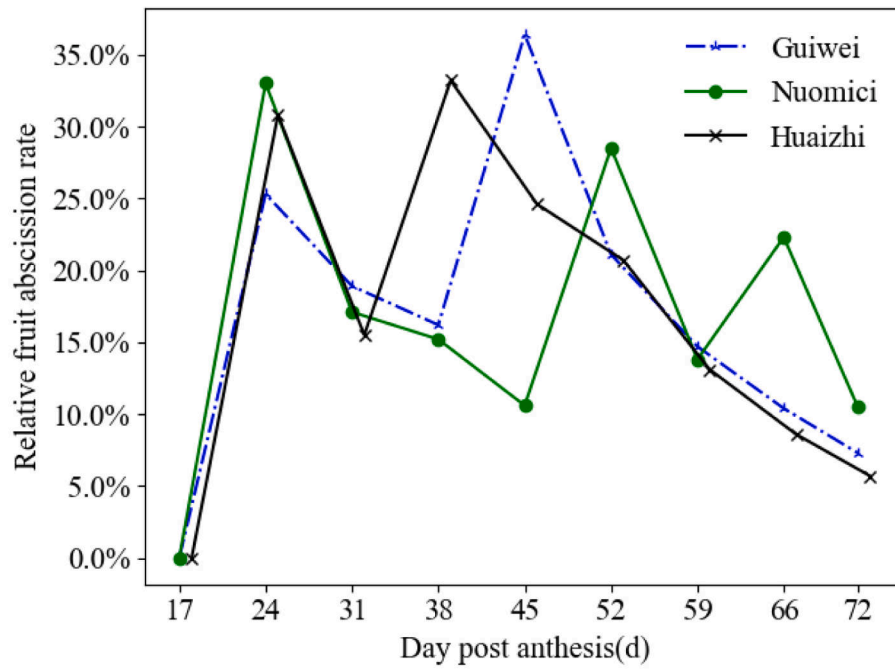
$$R = \frac{TP}{TP + FN} \quad (3)$$

$$AP(k) = \int_0^1 P(R) dR \quad (4)$$

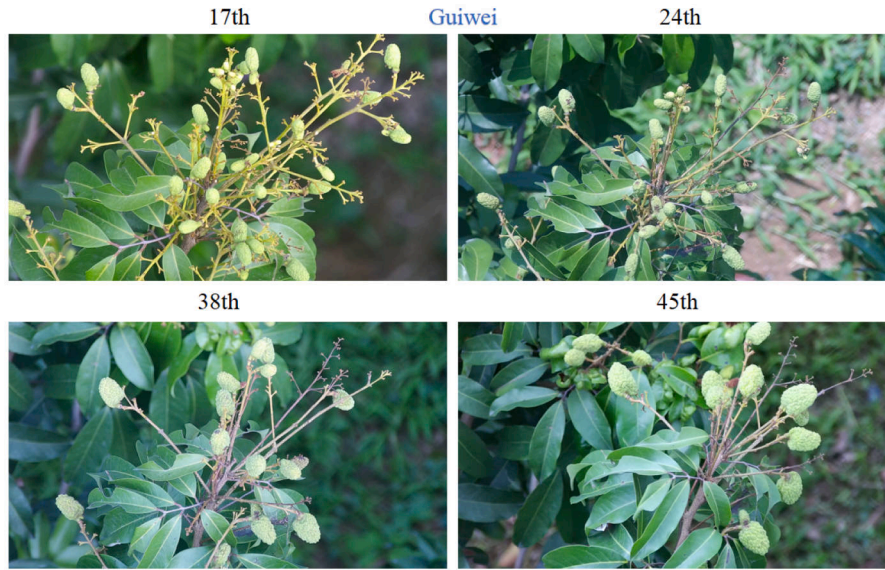
$$mAP = \frac{\sum_{k=1}^Q AP(k)}{Q} \quad (5)$$

where TP represents the number of correctly detected targets (true positive), FP represents the number of false detections (false positive), FN represents the number of missed detections (false negative), and *Q* represents the total number of categories. In this study, *Q* = 2; *AP(k)* represents the AP value of the *k*-th category.

To assess the effectiveness of the proposed method, this research randomly selected five branches from the 80 branches, and computed the Pearson Correlation Coefficient (PCC) between the ground truth and



(a) The relative fruit abscission of Guiwei, Huaizhi, and Nuomici.



(b)

Fig. 18. The illustration of relative fruit abscission. **a.** The relative fruit abscission of three varieties. **b, c, d.** The phenological states of the fruits of Guiwei, Huaizhi, and Nuomici during different fruit abscission peaks.

the prediction of fruits per branch per day over a span of 56 days.

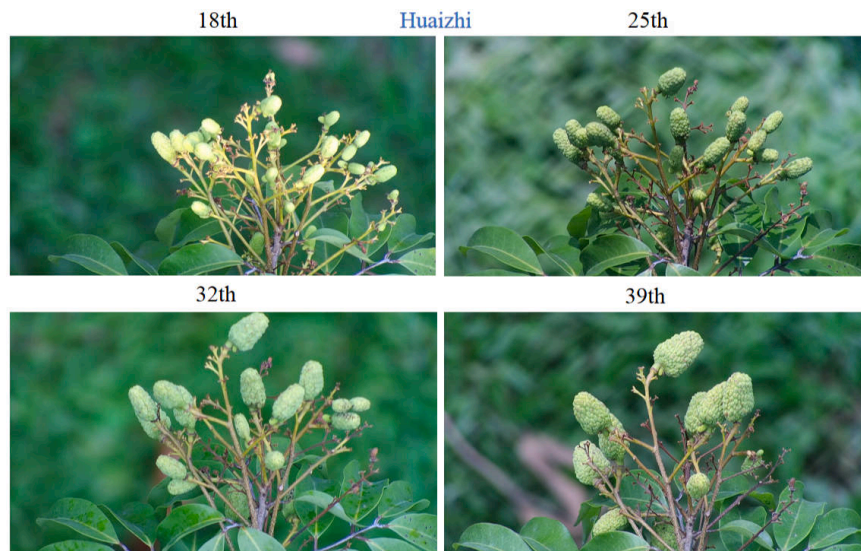
The formula for calculating the relative fruit abscission rate is as follows (Fan et al., 2015; Yan et al., 2019):

$$\text{relativefruitabscissionrate} = \frac{X_{i-1} - X_i}{X_{i-1}} \times 100\% \quad (6)$$

where, X_{i-1} represents the sum of fruit quantities on all branches of the variety on the $(i-1)$ -th day, and X_i represents the sum of fruit quantities on all branches of the variety on the i -th day.

3.3. Results of branch image matching experiment

Table 2 presents the results of the SE2-SuperGlue network on the test set of 676 images. Traditional feature matching algorithms, including SIFT, SURF, and ORB, were used for comparison. SIFT, SURF, and ORB are traditional methods used for extracting and describing image features. The algorithms are designed to extract local features from images and describe them, facilitating matching and recognition across different images. Fig. 13 visually compares SE2-SuperGlue with these



(c)



(d)

Fig. 18. (continued).

traditional algorithms, demonstrating that SE2-SuperGlue achieves a higher number of matched feature points. Fig. 14 illustrates the performance of SuperGlue and SE2-SuperGlue under varying degrees of rotation. While SuperGlue exhibits degraded performance under large rotation changes, SE2-SuperGlue remains robust, even with inplane rotation changes of 20°, 30° and 45°, respectively (Bökmán et al., 2022). This improvement highlights SE2-SuperGlue's enhanced rotational robustness. The differences between the algorithms are primarily observed in the matching score and processing speed. Results in Table 2 indicate that SE2-SuperGlue-based matching algorithms outperforms both SuperGlue and traditional feature matching algorithms in terms of the number of correctly matched feature points, while maintaining a competitive average processing time per frame. Specifically, the SE2-SuperGlue algorithm improves the matching score by 0.07, 0.32, 0.26, and 0.18 compared to SuperGlue (0.46), ORB (0.14), SURF (0.20), and SIFT (0.28), respectively. With a matching score of 0.53 and a speed of 177 ms/frame, the proposed branch image matching algorithm demonstrates superior accuracy and efficiency.

Fig. 15 shows the visualisation of the SE2-SuperGlue matching of results for two branches. Consequently, ROI-B sequence was obtained for all 80 branches throughout the continuous 56-day period. Subsequently, the 56-frame ROI-B of each branch were fed into a trained

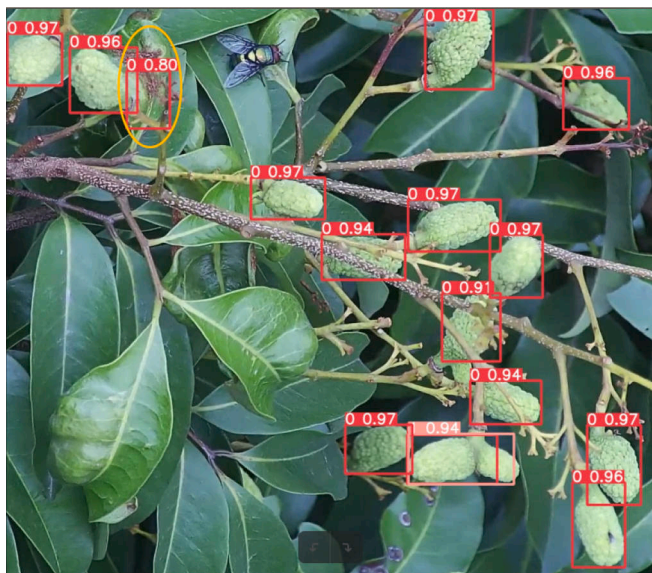
YOLOv8 model to detect and count the total number of fruits in class 1 and class 2 for each branch on a daily basis.

3.4. Results of fruit detection experiment

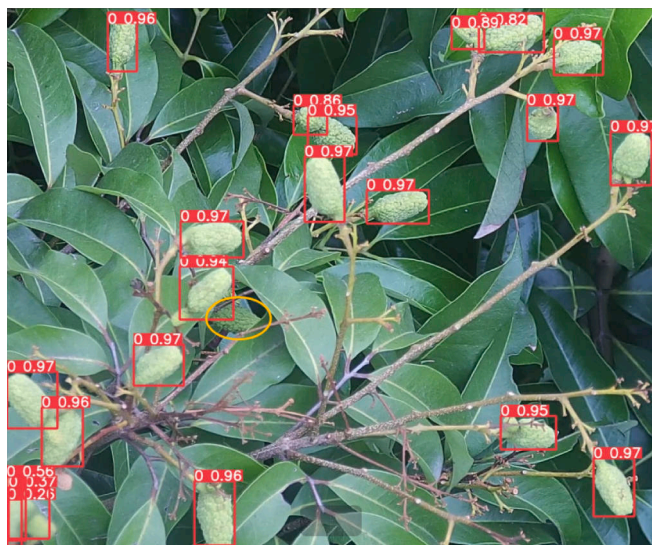
The comparative results are presented in Table 3. The YOLOv8n is compared with several commonly used algorithms known for their excellent performance in object detection. The YOLOv8n model improves the recognition result of the litchi fruit by 12 %, 9 %, 7.7 %, and 0.4 % compared to the YOLOv3-tiny model (86.7 %), YOLOXs model (89.7 %), Faster R-CNN model (91.0 %), and YOLOv5s model (98.3 %) respectively. These results highlight the advantages of the YOLOv8 model, including high accuracy, fast inference speed, and a compact model size.

3.5. Statistical results of litchi fruit abscission

The curves of the prediction and the ground truth are shown in Fig. 16. As evident from the figure, the average correlation coefficient R^2 is 0.98, indicating a close alignment between the model's predicted values and the ground truth from the images. This suggests that the proposed algorithm exhibits strong robustness and high accuracy even



(a) False detection



(b) Missed detection

Fig. 19. Failures of the YOLOv8 model.

in complex orchard environments.

The proposed algorithm was utilized to daily count the number of fruits for 56 days of 80 branches. Subsequently, the distribution of fruit set across all branches of three varieties and the relative fruit abscission rate for Guiwei, Nuomici, and Huaizhi were calculated, as illustrated in Fig. 17 and Fig. 18.

The litchi fruit abscission varies between varieties. As depicted in Fig. 17, significant disparities exist in the distribution of fruit set quantity among branches of different varieties. Fig. 18 shows that Guiwei reaches fruit abscission peaks on 17th-24th days and 38th-45th days after flowering, Huaizhi reaches fruit abscission peaks on 18th-25th and 32th-39th days after flowering, and Nuomici reaches fruit abscission peaks on 17th-24th days and 45th-52th days after flowering. In addition, Nuomici reaches the third fruit abscission peak on 59th-66th days after flowering before harvest. Litchi fruits experience two or three peak periods of fruit abscission before reaching maturity. Guiwei and Nuomici only have two peak periods of fruit abscission, while there is a third peak

of fruit abscission for Nuomici a few days before harvest (Hu et al., 2020; Ma et al., 2021). The final fruit abscission rates for Guiwei, Huaizhi, and Nuomici, are 94.7 %, 92.8 %, and 96.5 %, respectively, which is in line with the phenomenon of severe fruit abscission in litchi.

4. Discussion

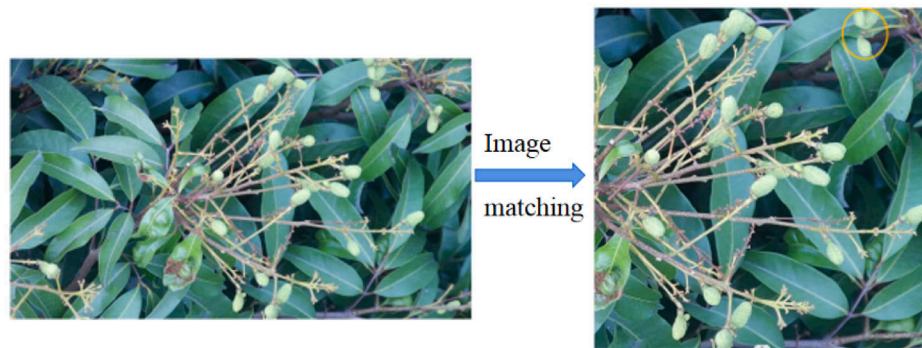
A novel and robust method for automatic tracking the litchi fruit abscission rate in complex natural orchard environments was proposed in this study. An image matching method based on SuperGlue was employed to determine the region of a single branch and extract the ROI-B from the time series images. The morphology of the same branch exhibited similarities over a period of growth, which justified the use of image matching. However, branches may undergo movements and rotations due to factors such as wind or growth-induced bending, necessitating enhancements to the SuperGlue algorithm. We tackled this problem by replacing the backbone CNN with a steerable CNN, ensuring equivariance to translations and rotations, contributing to a more robust model named SE2-SuperGlue. Considering the significant morphological changes of branches over the 56-day period, we divided the process into three distinct time periods for image matching. Although image matching successfully identified the ROI-B, instances occurred where other branches entered into this ROI-B, causing errors (See Fig. 20 a).

This research employed YOLOv8 to detect litchi fruits, achieving optimal accuracy, inference speed, and model size compared to existing object detection algorithms. However, there are also limitations (See Fig. 19). One limitation is misidentifying leaves as litchi due to their similar shapes, as is indicated by the yellow circle in Fig. 19 (a). The second limitation arises from the dense clustering of litchi fruits on the tree, which leads to occlusion of some fruit, as is highlighted by the yellow circle in Fig. 19 (b).

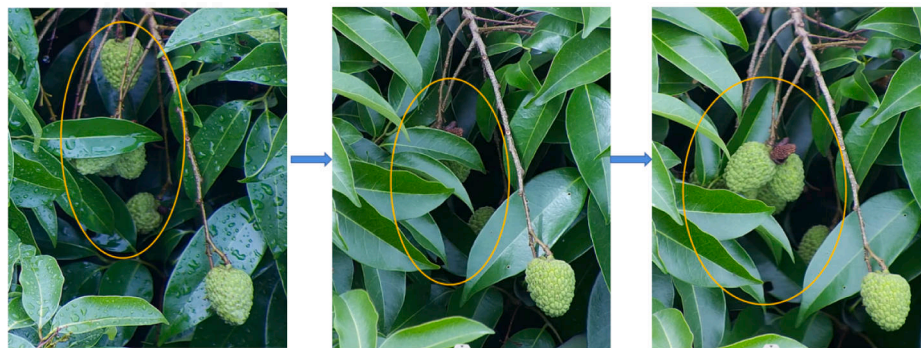
Utilizing the aforementioned methods, the automatic tracking and statistical analysis of litchi fruit abscission were conducted. As are depicted in Fig. 17 and Fig. 18, the fruit abscission rate of the three varieties is severe, with an increasing trend during fruit development, consistent with previous research on litchi fruit abscission (Fan et al., 2015; Cai et al., 2017). During the fruit abscission peak period (See Fig. 18), monitoring the phenological status of litchi fruits can guide the implementation of specific fruit preservation measures to stabilize litchi yield. The mean value of daily fruit set for the three varieties does not exhibit a continuous decrease over time (See Fig. 17), which can be attributed to the factors illustrated in Fig. 20. The first scenario arises from the close interconnections between branches during growth, leading to the inclusion of adjacent branches during image matching, as is depicted in the yellow circle in Fig. 20 (a). The second scenario is due to the substantial morphological changes in leaves and branches throughout growth, which may lead to fruit occlusion by leaves, as is shown in the yellow circle in Fig. 20 (b).

5. Conclusions

This paper introduced a novel approach that integrated our improved image matching and object detection for the automated tracking litchi fruit abscission in natural environments. A fixed-point and fixed-focus data acquisition system was developed for long-term tracking and imaging of litchi branch in the orchard. The improved SuperGlue network was employed to match branches and extract the ROI-B of the target branch from the time series images. Experimental results demonstrated that the SE2-SuperGlue network exhibited greater robustness to rotations and outperformed traditional feature matching algorithms in terms of matching scores, while also maintaining a powerful inference speed, with a score of 0.53 and an inference speed of 177 ms/frame, respectively. YOLOv8 achieved an mAP@0.5 score of 98.7 % on the test set. The mAP results suggested that YOLOv8 model surpassed YOLOv3-tiny, YOLOXs, Faster R-CNN, and YOLOv5s. Moreover, the fruit abscission on a total of 80 branches for three litchi



(a) Algorithm matched the fruits of adjacent branch



(b) Leaves occluded fruits

Fig. 20. The illustration of abnormal fruit number. a: algorithm matched the fruits of adjacent branch, b: leaves occluded fruits.

varieties, Guiwei, Nuomici, and Huaizhi, over a continuous 56-day period was analysed by using the proposed method. The average correlation coefficient R^2 between the prediction and the ground truth was 0.98, demonstrating the accurate detection of the proposed method. The analysis results revealed that the fruit abscission of the three varieties was severe, and the relative fruit abscission rate increased during fruit development. The proposed method could automatically monitor the dynamics of fruit abscission and establish the images of fruit state at the peaks of fruit abscission, providing valuable references for production management. Litchi fruit abscission is influenced by various factors, and in future research, we aim to explore the relationship between the growth and development of litchi at different phenological stages and fruit abscission.

CRediT authorship contribution statement

Tong Huang: Writing – original draft, Visualization, Methodology. **Jingfeng Guo:** Software, Investigation. **Long Yu:** Software. **Houbin Chen:** Writing – review & editing. **Zuanxian Su:** Writing – review & editing. **Yueju Xue:** Writing – review & editing, Supervision, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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